Canakinumab as Adjuvant Therapy in Patients With Completely Resected Non–Small-Cell Lung Cancer: Results From the CANOPY-A Double-Blind, Randomized Clinical Trial

Authors: Edward B. Garon, MD , Shun Lu, MD , Yasushi Goto, MD, PhD , Pedro De Marchi, MD, PhD , Luis Paz-Ares, MD, PhD , David R. Spigel, MD , Michael Thomas, MD , ... SHOW ALL ... , and Byoung Chul Cho, MD, PhD AUTHORS INFO & AFFILIATIONS

J Clin Oncol **42**, 180-191(2024) Volume 42, Number 2 DOI: 10.1200/JCO.23.00910

4,711 / 53 score 26

PDF

Abstract

Purpose

Effective treatments for resectable non–small-cell lung cancer (NSCLC) are limited and relapse rates are high. The interleukin (IL)-1β pathway has been linked with tumor development and progression, including in the Canakinumab Anti-Inflammatory Thrombosis Outcomes cardiovascular study in which IL-1β pathway inhibition with canakinumab reduced lung cancer incidence and mortality in an exploratory analysis.

Methods

CANOPY-A (ClinicalTrials.gov identifier: NCT03447769) is a phase III, randomized, double-blind, multicenter study of canakinumab versus placebo for adult patients with stage II-IIIA or IIIB (T >5 cm, N2-positives II-IIIB; American Joint Committee on Cancer/Union for International Cancer Control version 8), completely resected NSCLC who had received adjuvant cisplatin-based chemotherapy. The primary end point was disease-free survival (DFS) and the key secondary end point was overall survival (OS).

Results

In total, 1,382 patients were randomized to 200 mg canakinumab (n = 693) or placebo (n = 689) once every 3 weeks for 18 cycles. Grade ≥3 adverse events (AEs) were reported in 20.8% and 19.6% of patients receiving canakinumab and placebo, respectively; AEs led to discontinuation in 4.3% and 4.1% of patients in these groups, respectively. This study did not meet its primary end point. Median DFS was 35.0 months (canakinumab arm) and 29.7 months (placebo arm; hazard ratio, 0.94; 95% CI, 0.78 to 1.14; one-sided P = .258). DFS subgroup analyses did not show any meaningful differences between arms. As expected, because of canakinumab-driven IL-1β pathway inhibition, C-reactive protein and IL-6 levels decreased in the canakinumab arm versus placebo arm, but had no correlation with differential clinical outcomes. OS was not formally tested as DFS was not statistically significant.

Conclusion

CANOPY-A did not show a DFS benefit of adding canakinumab after surgery and adjuvant cisplatin-based chemotherapy in patients with resected, stage II-III NSCLC. No new safety signals were identified with canakinumab.

Introduction

Lung cancer is the leading cause of cancer-related mortality worldwide.[1] In the United States, non–small-cell lung cancer (NSCLC) accounts for approximately 85% of all lung cancer cases.[2] Currently, the standard of care for patients with resectable and operable, stage II-III NSCLC is surgical resection combined with adjuvant cisplatin-based chemotherapy with or without immunotherapy.[3,4] For patients with epidermal growth factor receptor (*EGFR*)–mutated tumors, adjuvant treatment with the targeted therapy osimertinib is used[5]; however, relapse after adjuvant treatment remains common, and new systemic treatments are needed.[6-8]

Context

Key Objective

Does treatment with the interleukin (IL)-1β inhibitor canakinumab improve disease-free survival (DFS) in patients with resectable, early-stage non–small-cell lung cancer (NSCLC) after surgery and adjuvant cisplatin-based chemotherapy?

Knowledge Generated

No DFS benefit was observed in patients receiving canakinumab in the postsurgery, adjuvant setting. Median DFS lengths were 35.0 months (canakinumab arm; n = 693) and 29.7 months (placebo arm; n = 689; hazard ratio, 0.94; 95% CI, 0.78 to 1.14; one-sided *P* = .258). DFS subgroup analyses did not show any meaningful differences between arms. No new safety signals were observed after canakinumab treatment.

Relevance *(T.E. Stinchcombe)*

Canakinumab did not demonstrate activity in resected NSCLC. Targeting IL-1β continues to be of interest for preclinical studies and drug development.*

*Relevance section written by *JCO* Associate Editor Thomas E. Stinchcombe, MD.

Inflammation is associated with cancer development and progression, and reducing inflammation represents a potential strategy for the prevention and treatment of cancer.[9] Certain risk factors, including smoking and air pollution, cause chronic inflammation, which can increase the risk of lung cancer development.[10] Interleukin (IL)-1β, a proinflammatory cytokine, has been shown to play a role in tumorigenesis.[11] Elevated levels of IL-1β and its downstream effectors (eg, C-reactive protein [CRP] and IL-6) have been linked with lung cancer progression.[9,12] Inhibition of IL-1β has been associated with suppressed tumor progression and enhanced tumor immunity[11,13]; however, targeting inflammation in lung cancer has not been therapeutically harnessed.[9]

Canakinumab is a high-affinity monoclonal antibody, blocking IL-1β interaction with its receptor, and is approved for treating autoinflammatory conditions.[14] The phase III Canakinumab Anti-Inflammatory Thrombosis Outcomes Study (CANTOS; ClinicalTrials.gov identifier: NCT01327846) investigated canakinumab for the secondary prevention of major adverse cardiovascular events in patients with atherosclerosis, a history of myocardial infarction, and elevated CRP.[15] An exploratory analysis of CANTOS demonstrated dose-dependent reductions in lung cancer incidence and mortality with canakinumab compared with placebo.[16] Further analysis from CANTOS showed that among patients diagnosed with lung cancer during the trial, 43% had detectable circulating tumor DNA (ctDNA) with annotated variant(s) according to the Catalogue of Somatic Mutations in Cancer at baseline, suggesting that some patients in CANTOS had undiagnosed lung cancer at the study start.[17]

The CANOPY program was designed to further investigate the potential role of canakinumab in the treatment of NSCLC. The CANOPY-A (ClinicalTrials.gov identifier: NCT03447769) study compared adjuvant canakinumab versus placebo in patients with NSCLC.

Methods

Study Design, Treatment, and Objectives

CANOPY-A is a phase III, multicenter, randomized, double-blind, placebo-controlled study

evaluating the efficacy and safety of canakinumab versus placebo as adjuvant therapy in adult patients with early-stage, resected NSCLC.

Patients enrolled between April 2018 and December 2021 and were randomly assigned 1:1 (permuted block random assignment) to receive subcutaneous canakinumab 200 mg or placebo once every 3 weeks for 18 cycles (Figs 1 and 2). Patients were stratified by stage (IIA *v* IIB *v* IIIA *v* IIIB with T >5 cm, N2-positive disease), as defined according to the American Joint Committee on Cancer/Union for International Cancer Control (AJCC/UICC) version 8 stage; histology (squamous *v* nonsquamous); and geographical region (western Europe and North America *v* eastern Asia *v* rest of the world).

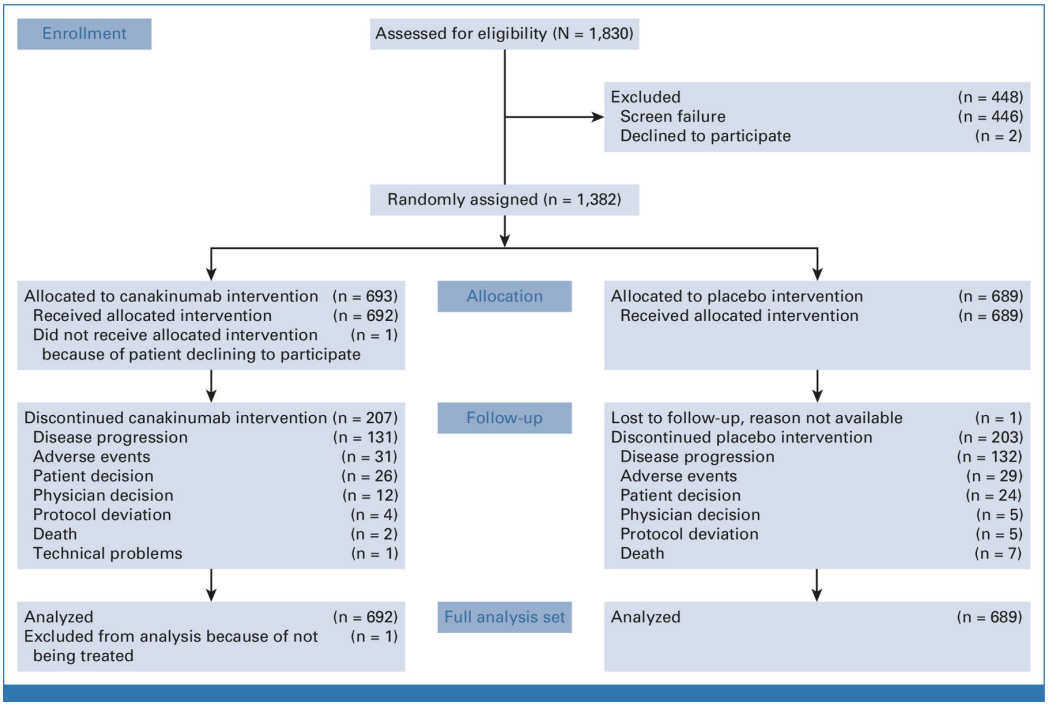


Fig 1. CANOPY-A CONSORT diagram.

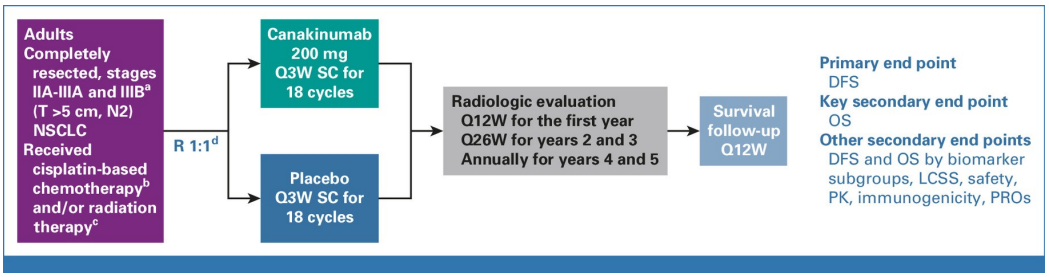


Fig 2. CANOPY-A study design. [a]Stages were defined using the AJCC/UICC version 8 staging recommendation. [b]Mandatory, except for patients with stage IIA disease with no nodal involvement. [c]If indicated per local guidelines/practice. [d]Randomized according to stratification factors (AJCC/UICC stage, histology, and region). AJCC, American Joint Committee on Cancer; DFS, disease-free survival; LCSS, lung cancer–specific survival; N, node; NSCLC, non–small-cell lung cancer; OS, overall survival; PK, pharmacokinetics; PRO, patient-reported outcome; Q3W, once every 3 weeks; Q12W, once every 12 weeks; Q26W, once every 26 weeks; R, randomization; SC, subcutaneous; T, tumor; UICC, Union for International Cancer Control.

Patients received treatment until completion (18 cycles), NSCLC disease recurrence determined by the investigator, unacceptable toxicity, investigator's decision to discontinue treatment, start of a

new antineoplastic therapy, death, loss to follow-up, or consent withdrawal.

The primary end point was disease-free survival (DFS), assessed by investigator per RECIST version 1.1. The key secondary end point was overall survival (OS), to be tested if DFS was statistically significant. Other secondary end points included DFS and OS by biomarker subgroups, lung cancer–specific survival (LCSS), safety, pharmacokinetics, and immunogenicity. Details of LCSS methods and a supplementary analysis to estimate the treatment effects on the basis of DFS had the COVID-19 pandemic not occurred are provided in Appendix 1 (online only).

Patient Population

Adult patients with completely resected, AJCC/UICC version 8 stage II-IIIA or IIIB (T >5 cm, N2-positive disease) NSCLC were recruited. Patients had received and recovered from at least two cycles of cisplatin-based therapy within 60 days of surgery (except for patients with stage IIA disease with no nodal involvement, unless recommended by the treating physician) and radiation therapy if indicated per local practice. Eligible patients had adequate organ function and an Eastern Cooperative Oncology Group performance status of 0 or 1. Patients were eligible regardless of driver mutation status. Patients with unresectable or metastatic disease, malignant disease other than NSCLC, or lung interstitial disease grade ≥2, and those who had received neoadjuvant therapy, were ineligible.

Assessments

Imaging (computed tomography [CT] and/or magnetic resonance imaging) was performed at screening/baseline, once every 12 weeks during year 1, once every 26 weeks during years 2 and 3, and annually during years 4 and 5. First disease recurrence was detected by CT scan and a biopsy was performed in cases of nonconclusive radiologic evidence. Patients who discontinued treatment were followed up once every 12 weeks for survival until the final OS analysis or until death, loss to follow-up, or consent withdrawal for survival follow-up.

Safety was monitored. All adverse events (AEs) were reported for at least 130 days after the last administration of canakinumab and graded according to the Common Terminology Criteria for AEs version 5.

Biomarker analyses were conducted using tumor tissue and blood samples. Collection of archival, resected tumor samples during screening was mandatory. Additional tumor samples (core, excisional, and incisional biopsies) were collected at the time of disease recurrence. Correlative analyses reported focus on the baseline tissue collection time point only, because of availability from most patients, and depended on testing priority, tissue quality, and local regulations. Collection of blood samples for circulating biomarker analyses (eg, IL-6 and CRP) was mandatory at baseline, on treatment, at post-treatment follow-up, and at disease progression. Detailed biomarker assessments are described in Appendix 1.

Statistical Analysis

The full analysis set comprised all patients assigned to study treatment by random assignment. DFS of canakinumab versus placebo arms was analyzed using a stratified log-rank test at an overall one-sided 2.5% significance level. This study had a two-look (Lan-DeMets) group sequential design with an O'Brien-Fleming type α-spending function. A user-defined γ function (with γ = −3) was

used as a β-spending function to define a nonbinding futility rule at the interim analysis of DFS.

The DFS distribution was estimated using the Kaplan-Meier method, and the sample size calculation was based on the primary variable DFS. To ensure 90% power to detect a hazard ratio (HR) of 0.716, a total of 392 DFS events needed to be observed. Further details of the sample size and interim analysis are available in Appendix 1.

The secondary end point, OS, would be tested if DFS was statistically significant. LCSS distribution was estimated using the Kaplan-Meier method; HRs and 95% CIs were calculated using a stratified Cox model.

Ethical Oversight

This study (ClinicalTrials.gov identifier: NCT03447769) was conducted in accordance with the International Council for Harmonisation of Technical Requirements for Registration of Pharmaceuticals for Human Use Harmonized Tripartite Guidelines for Good Clinical Practice, with applicable local regulations, including the European Directive 2001/20/EC and US Code of Federal Regulations Title 21, and the Declaration of Helsinki. The Protocol (online only) and all amendments were reviewed and approved by an institutional review board/independent ethics committee/research ethics board before study start. All patients provided written informed consent before study start.

Results

Patient Characteristics and Disposition

In total, 1,382 patients were randomly assigned to receive canakinumab (n = 693) or placebo (n = 689). Median time from random assignment to data cutoff (March 17, 2022) was 18.9 months (range, 2.8-47.2). Baseline patient and disease characteristics were well balanced between arms (Table 1).

TABLE 1. Patient Baseline Characteristics

Demographic Variable	Canakinumab Group (n = 693)	Placebo Group (n = 6
Age, years, median (range)	63.0 (27-83)	62.0 (21-82)
Sex, No. (%)		
Male	430 (62.0)	432 (62.7)
Female	263 (38.0)	257 (37.3)
ECOG performance status, No. (%)		
0	444 (64.1)	449 (65.2)
1	249 (35.9)	240 (34.8)
Region, No. (%)		
East Asia	237 (34.2)	233 (33.8)
Rest of the world	176 (25.4)	177 (25.7)
Western Europe and North America	280 (40.4)	279 (40.5)
Smoking history, No. (%)		
Never smoker	183 (26.4)	180 (26.1)
Former smoker	425 (61.3)	419 (60.8)
Current smoker	85 (12.3)	90 (13.1)
Time from diagnosis to randomization, months, median (range)	5.8 (0.5-37.9)	5.8 (0.3-34.2)
Tumor histology/cytology, No. (%)		
Nonsquamous cell carcinoma	519 (74.9)	507 (73.6)
Squamous cell carcinoma	174 (25.1)	182 (26.4)
Stage,[a] No. (%)		
IB	2 (0.3)	1 (0.1)
IIA	114 (16.5)	115 (16.7)
IIB	264 (38.1)	259 (37.6)
IIIA	271 (39.1)	271 (39.3)
IIIB	42 (6.1)	43 (6.2)
Type of previous therapy, No. (%)		
Chemotherapy	636 (91.8)	620 (90.0)
Radiotherapy	84 (12.1)	77 (11.2)
PD-L1 category,[b] No. (%)		
<1%	211 (30.4)	203 (29.5)
1%-49%	99 (14.3)	119 (17.3)
≥50%	86 (12.4)	96 (13.9)
Missing[c]	297 (42.9)	271 (39.3)

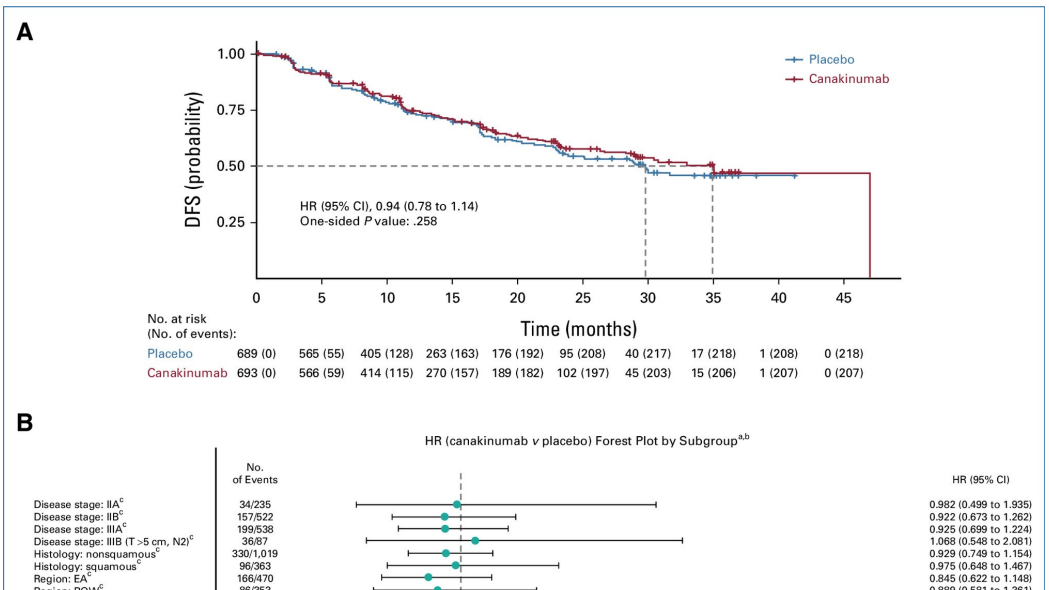
CD8 category, No. (%)		
<Median	213 (30.7)	225 (32.7)
≥Median	216 (31.2)	224 (32.5)
Missing[c]	264 (38.1)	240 (34.8)
CRP category, No. (%)		
<2 mg/L	336 (48.5)	343 (49.8)
≥2 mg/L	297 (42.9)	302 (43.8)
Missing[c]	60 (8.7)	44 (6.4)
EGFR mutation status,[d] No. (%)		
Positive	50 (7.2)	60 (8.7)
Negative	189 (27.3)	172 (25.0)
Unknown	2 (0.3)	4 (0.6)
Missing[c]	452 (65.2)	453 (65.7)
ALK rearrangement status,[d] No. (%)		
Positive	9 (1.3)	14 (2.0)
Negative	239 (34.5)	198 (28.7)
Unknown	2 (0.3)	1 (0.1)
Missing[c]	443 (63.9)	476 (69.1)

Abbreviations: *ALK*, anaplastic lymphoma kinase; CD8, cluster of differentiation 8; CRP, C-reactive protein; ECOG, Eastern Cooperative Oncol
Group; eCRF, electronic Case Report Form; *EGFR*, epidermal growth factor receptor; TMN, tumor, metastasis, node.
[a]Derived based on TMN; stages based on the eCRF.
[b]Excluding patients with PD-L1 level data missing: PD-L1 <1%: 53.3% (canakinumab) and 48.6% (placebo); PD-L1 1%-49%: 25.0% (canakinumab)
28.5% (placebo); PD-L1 ≥50%: 21.7% (canakinumab) and 23.0% (placebo).
[c]Biomarker results could be missing because of insufficient quantity or quality (including tissue sample age) for biomarker testing or because
local regulations.
[d]Local assessment, testing not required.

At data cutoff, 337 (48.6%) and 349 (50.7%) patients had completed treatment in the canakinumab and placebo arms, respectively. Treatment was discontinued before completion in 207 (29.9%) and 203 (29.5%) patients in the canakinumab and placebo arms, respectively (Appendix Table A1). The main reasons for discontinuation were progressive disease (18.9% and 19.2% in the canakinumab and placebo arms, respectively), AEs (4.5% and 4.2% in the canakinumab and placebo arms, respectively), and patient decision (3.8% and 3.5% in the canakinumab and placebo arms, respectively; Appendix Table A1). Median duration of exposure to study treatment was similar between arms (11.7 months [quartile 1 to quartile 3: 6.2-12.4] and 11.8 months [quartile 1 to quartile 3: 6.2-12.4] in the canakinumab and placebo arms, respectively).

Efficacy

The CANOPY-A phase III study did not meet its primary end point of DFS (Fig 3A). In total, 208 (30.0%) events were reported in the canakinumab arm and 218 (31.6%) in the placebo arm. Median DFS was 35.0 months and 29.7 months in the canakinumab and placebo arms, respectively (HR, 0.94; 95% CI, 0.78 to 1.14; one-sided P = .258). Subgroup analyses of DFS, on the basis of demographics, baseline disease characteristics, and biomarkers of interest, did not show any meaningful differences between treatment arms (Fig 3B).



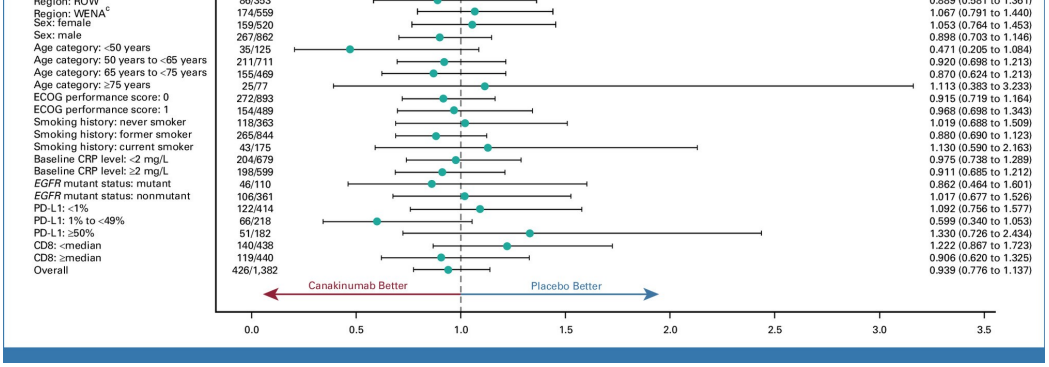
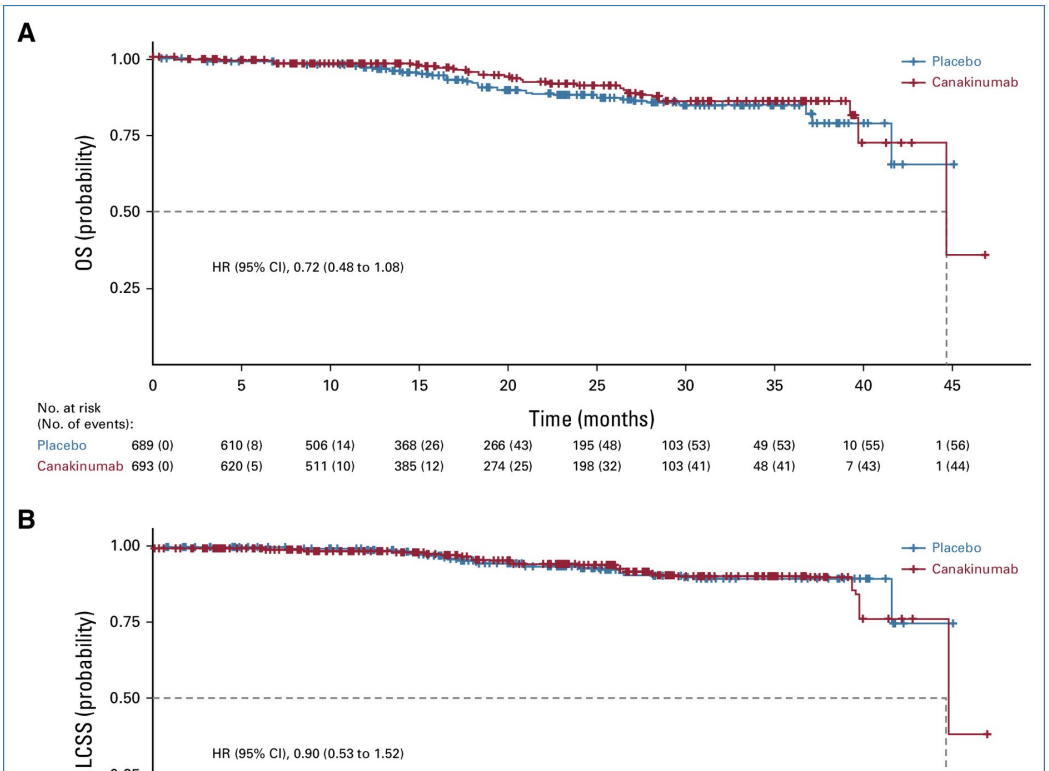


Fig 3. Primary end point of DFS with canakinumab versus placebo. (A) DFS analysis. (B) DFS subgroup analysis. [a]Patients with data missing for *EGFR* mutation status and PD-L1, CD8, and CRP levels have been removed. [b]The baseline characteristics of patients with biomarker data available were not representative of the full analysis set for PD-L1, CD8, and *EGFR* but were representative for CRP. [c]By IRT. CD8, cluster of differentiation 8; CRP, C-reactive protein; DFS, disease-free survival; EA, eastern Asia; ECOG, Eastern Cooperative Oncology Group; *EGFR*, epidermal growth factor receptor; HR, hazard ratio; IRT, Interactive Response Technology; N, node; ROW, rest of the world; T, tumor; WENA, western Europe and North America.

In light of the COVID-19 pandemic and its potential impacts on treatment effects, a supplementary analysis censored patients who died or discontinued treatment due to COVID-19, or who started medication for suspected/confirmed COVID-19. On the basis of this analysis, there was no treatment effect (HR, 0.96; 95% CI, 0.79 to 1.16).

The key secondary end point, OS, was not formally tested as DFS was not statistically significant (OS HR, 0.72; 95% CI, 0.48 to 1.08; Fig 4A). Only 100 events were reported: 44 (6.3%) in the canakinumab arm and 56 (8.1%) in the placebo arm. LCSS (defined as the time from the date of random assignment to the date of death due to lung cancer) was similar between arms (HR, 0.90; 95% CI, 0.53 to 1.52; Fig 4B). Overall, 60 LCSS events were observed, including 30 events in the canakinumab arm and 30 events in the placebo arm. Fourteen patients on canakinumab and 26 patients on placebo died due to other reasons and were censored for the LCSS analysis.



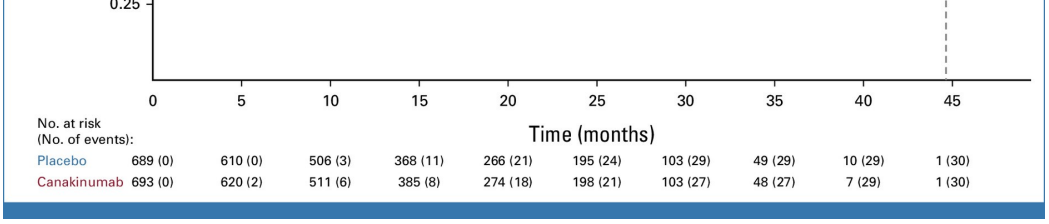


Fig 4. (A) OS and (B) LCSS in the canakinumab and placebo arms. HR, hazard ratio; LCSS, lung cancer–specific survival; OS, overall survival.

Safety

Regardless of causality, only two AEs of any grade were reported in ≥10% of patients (Appendix Table A2): both were less frequent in the canakinumab arm compared with the placebo arm (cough [11.4% *v* 14.8%] and arthralgia [10.0% *v* 12.0%]). Grade 3/4 AEs were experienced by 139 (20.1%) and 123 (17.9%) patients in the canakinumab and placebo arms, respectively, and grade 5 AEs were experienced by five (0.7%) and 12 (1.7%) patients in these groups, respectively. Rates of AEs, serious AEs (SAEs), and fatal SAEs were similar between arms (Table 2).

TABLE 2. Overview of AEs

	Canakinumab Group (n = 692), No. (%)			Placebo Group (n = 689), No. (%)		
AE	Any Grade	Grade 3/4	Grade 5	Any Grade	Grade 3/4	Grade
All deaths	44 (6.4)	–	–	56 (8.1)	–	–
On-treatment deaths[a]	9 (1.3)	–	–	16 (2.3)	–	–
AEs	582 (84.1)	139 (20.1)	5 (0.7)	586 (85.1)	123 (17.9)	12 (1.
Treatment-related	269 (38.9)	44 (6.4)	0	217 (31.5)	20 (2.9)	2 (0.
SAEs	112 (16.2)	54 (7.8)	5 (0.7)	122 (17.7)	56 (8.1)	12 (1.
Treatment-related	13 (1.9)	9 (1.3)	0	12 (1.7)	7 (1.0)	2 (0.
Fatal SAEs	5 (0.7)	0	5 (0.7)	12 (1.7)	0	12 (1.
Treatment-related	0	0	0	2 (0.3)	0	2 (0.
AEs leading to discontinuation	30 (4.3)	17 (2.5)	1 (0.1)	28 (4.1)	18 (2.6)	0
Treatment-related	18 (2.6)	10 (1.4)	0	10 (1.5)	9 (1.3)	0
AEs leading to dose interruption	93 (13.4)	41 (5.9)	0	76 (11.0)	23 (3.3)	0
AEs requiring additional therapy	449 (64.9)	88 (12.7)	2 (0.3)	454 (65.9)	82 (11.9)	2 (0.

Abbreviations: AE, adverse event; SAE, serious AE.
[a]Defined as deaths that occurred from the date of first administration of study treatment to 130 days after the date of last actual administratior

Treatment-related AEs (TRAEs) occurred in 38.9% and 31.5% of patients in the canakinumab and placebo arms, respectively, of which 6.4% and 2.9% were grade 3/4. Only in the placebo arm were grade 5 TRAEs (0.3%) reported (Table 2).

Infections and neutropenia are expected AEs associated with canakinumab. All infections (reported by Medical Dictionary for Regulatory Activities version 25 system organ class) were reported with comparable incidences between treatment arms, with grade 3/4 events reported in 3.3% and 2.8%, and grade 5 events in 0.4% and 0.6%, of patients in the canakinumab and placebo arms, respectively (Appendix Table A3). Neutropenia incidence was higher in the canakinumab arm compared with the placebo arm (grade 3/4: 1.3% and 0.3%, respectively; no grade 5), as was decreased neutrophil count (grade 3/4: 1.7% and 0.3%, respectively; no grade 5; Appendix Table A2).

AEs leading to dose interruption occurred in 13.4% and 11.0% of patients in the canakinumab and placebo arms, respectively; 5.9% and 3.3%, respectively, were grade 3/4, and no grade 5 AEs leading to dose interruption were reported (Table 2). Any-grade TRAEs leading to discontinuation

occurred in 18 (2.6%) and 10 (1.5%) patients in the canakinumab and placebo arms, respectively.

On-treatment deaths, defined as deaths that occurred between the first day and 130 days after the last administration of study treatment, were lower in the canakinumab arm (n = 9: disease progression [n = 4], infections and infestations [n = 3], cardiovascular [n = 1], and respiratory [n = 1]) compared with the placebo arm (n = 16: disease progression [n = 4], infections and infestations [n = 4], cardiovascular [n = 4], respiratory [n = 2], injury [n = 1], and nervous system disorders [n = 1]; Table 2). Overall, there were 44 (6.4%) and 56 (8.1%) deaths in the canakinumab and placebo arms, respectively, including 30 (4.3%) deaths in the canakinumab arm and 30 (4.4%) deaths in the placebo arm due to disease progression (Appendix Table A4).

Biomarkers

There was limited evidence of systemic inflammation in patients at study entry: median baseline CRP level was 1.8 mg/L (quartile 1 to quartile 3: 0.8-4.3) and median baseline IL-6 level was 3.6 pg/L (quartile 1 to quartile 3: 2.2-6.7). Although CRP and IL-6 levels decreased from baseline in the canakinumab arm, as expected, little change was noted in the placebo arm. The mean reductions induced by canakinumab compared with placebo ranged from 43.9% to 51.9% for CRP and from 31.4% to 35.8% for IL-6 (Figs 5A and 5B). There was no benefit associated with canakinumab in patient subgroups defined by baseline CRP and IL-6 levels (Fig 3B, data not shown).

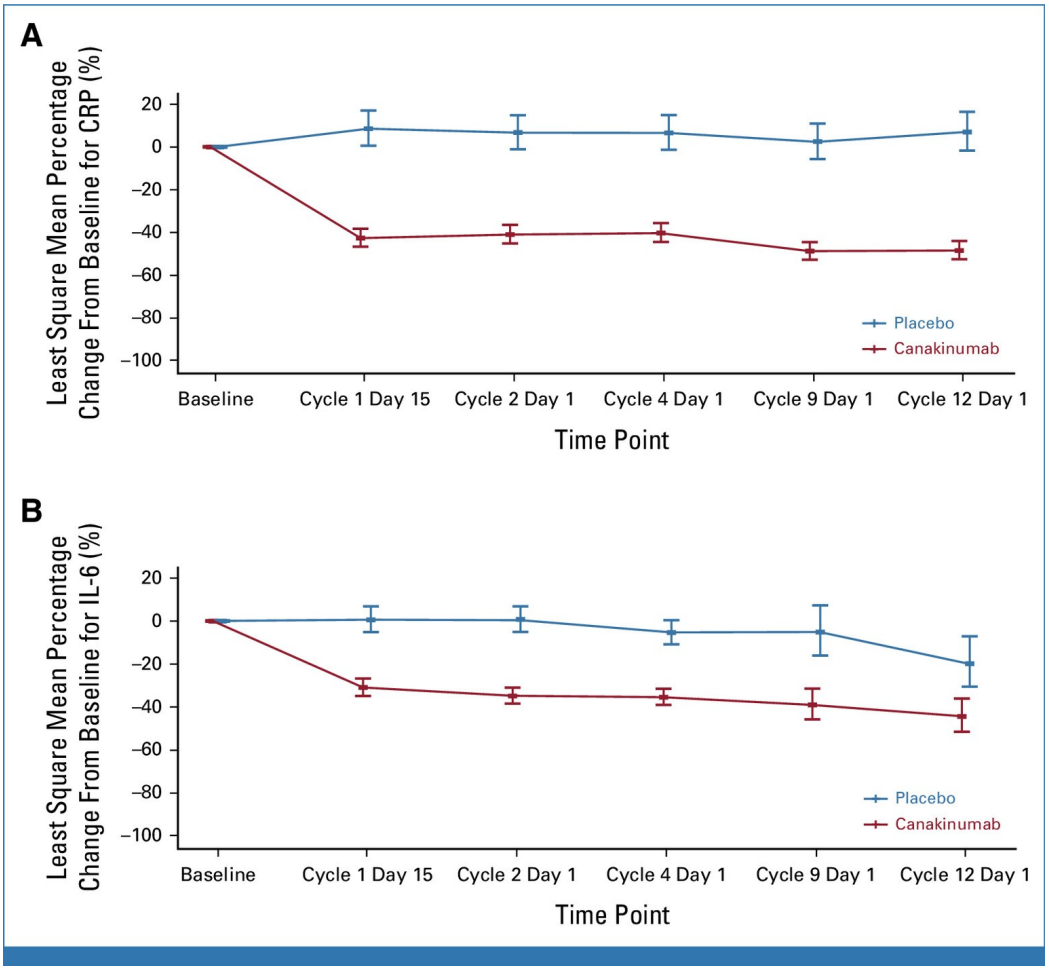


Fig 5. Least square mean percentage change from baseline in the canakinumab and placebo arms (A) in CRP and (B) in IL-6. CRP, C-reactive protein; IL, interleukin.

Discussion

Discussion

In an exploratory analysis of the CANTOS trial, reductions in lung cancer incidence and mortality were seen, which led to the hypothesis that canakinumab might improve clinical outcomes for patients with NSCLC. To test this, a suite of CANOPY clinical studies was opened to explore canakinumab treatment for NSCLC in various settings. Although the CANOPY-1 and CANOPY-2 (unpublished data) studies did not show improvements in progression-free survival or OS in either the advanced or metastatic settings in either trial, patient-reported outcomes from CANOPY-1 suggested beneficial effects of canakinumab on symptoms.[18-20] By contrast, the CANOPY-A study aimed to explore the efficacy of canakinumab in the early-stage disease setting in patients without evidence of disease after complete resection (followed by adjuvant chemotherapy and/or radiotherapy when applicable), a more relevant setting to test the hypothesis on the basis of CANTOS data. Of note, the CANOPY-N study investigated canakinumab as a single agent and in combination with a checkpoint inhibitor in the neoadjuvant setting in patients who were treatment-naïve. Decreases in inflammation markers and immunosuppressive features in the tumor microenvironment were observed with IL-1β inhibition, but there was minimal single-agent activity.[21]

In CANOPY-A, the primary end point of DFS was not met, and subgroup analyses of DFS, including by smoking history, were explored. In the CANTOS study, patients who developed lung cancer generally had a history of smoking, and the results suggested that canakinumab had a slightly stronger effect in current versus former smokers. This observation seemed consistent with the anticipated effects of canakinumab, as patients with a history of smoking would be expected to be more likely to develop inflammatory disease and, thus, be more likely to derive benefit from canakinumab treatment.[22] Yet, in CANOPY-A, there was no benefit of canakinumab in subgroups of patients defined by smoking status. Patients were enrolled in CANOPY-A from April 11, 2018, to December 23, 2021, during which time new data were released for checkpoint inhibitors in the adjuvant setting. We cannot exclude the possibility that the change in treatment pattern after the approval of immunotherapy for patients who are PD-L1–positive may have resulted in a larger PD-L1–low/–negative population in CANOPY-A; however, there was no clear trend across PD-L1 subgroups for canakinumab benefit. Additional CANOPY-A study limitations include the long time from diagnosis to random assignment, based, in part, on the need for chemotherapy in most patients and radiation therapy in some; short time from random assignment to data cutoff; and lack of pathologic confirmation of recurrent disease.

Although not formally tested, OS was numerically greater in the canakinumab arm compared with the placebo arm. Only 100 deaths were reported, and the separation of the Kaplan-Meier curves occurred after completion of therapy. Furthermore, LCSS rates were similar between the two treatment arms, with relatively more patient deaths, not from lung cancer, in the placebo arm, including cardiovascular disease–related fatalities.

Different factors may underpin the disparate results from CANTOS and CANOPY-A, such as the differences in the populations recruited: patients had cardiovascular disease and myocardial infarction versus patients with NSCLC, respectively. CRP level was higher in patients diagnosed with lung cancer in CANTOS than in patients in CANOPY-A (6.0 mg/L *v* 1.8 mg/L, respectively).[16] Nevertheless, the effect of canakinumab on CRP level in CANOPY-A was similar to that observed in the CANTOS trial, with a significant CRP reduction of up to 41%,[16] suggesting that canakinumab

effectively inhibited IL-1β signaling in CANOPY-A. Although individual doses in CANOPY-A were lower than in CANTOS, on the basis of considerably more frequent dosing, dose density was greater. Although CANTOS enrolled patients who were required to have a CRP level above 2 mg/L, no threshold was set for patients enrolled in CANOPY-A. As patients with lung cancer have been reported to exhibit increased levels of inflammation,[23,24] it was expected that the CANOPY-A study population would be enriched for elevated CRP without the need to set a threshold. Additionally, not setting a limit on baseline CRP level ensured further investigation into the relationship between CRP level and clinical outcomes with canakinumab would be possible, as well as that patients potentially capable of benefiting from IL-1β inhibition were not excluded. The baseline CRP levels in CANOPY-A were lower than anticipated and reflect a relatively low inflammatory status of patients with NSCLC after surgical resection and chemotherapy. There was no DFS benefit with canakinumab in CANOPY-A in the subgroup of patients who had a baseline CRP level above 2 mg/L. In CANOPY-1, there was a trend for OS benefit with canakinumab in the lowest quartile of CRP levels.[18] Taken together, baseline and decreases in CRP levels appear not to be relevant biomarkers for predicting or determining response to canakinumab for lung cancer relapse in the covered range of baseline CRP levels.[25] Further molecular analyses in CANOPY-A are currently ongoing and could help identify subgroups of patients who may benefit from canakinumab in the adjuvant setting.

The end points also differed between CANTOS and CANOPY-A. Lung cancer incidence was not a prespecified efficacy end point in CANTOS, and occurrence was derived from AE data. Thus, the strength of these data might have been overestimated. As the most common cause of cancer death is lung cancer, it would likely have been the only disease with the potential to show a reduction in cancer deaths. However, the analyses did not adjust for multiple hypothesis testing. The fact that one type (lung cancer) of one class of AEs (cancer) was analyzed without adjusting for multiplicity could explain why the exploratory evidence from CANTOS was not confirmed in this study.

In a retrospective analysis of the CANTOS trial, some patients diagnosed with lung cancer during treatment with canakinumab were determined to be positive for ctDNA at baseline and, therefore, likely had undiagnosed lung cancer at the start of treatment. This led to the assumption that canakinumab could prevent progression of micrometastatic disease, and avoid progression to clinically detectable disease and noticeable symptoms. It is important to note that the limited ctDNA data set prevented the possibility of making a firm conclusion on whether reduced lung cancer incidence and mortality after canakinumab treatment were due to reduced disease progression or prevention of lung cancer development over the course of the CANTOS trial (including patients with ctDNA-negative results at baseline). If canakinumab only has an impact on the prevention of lung cancer development, but not on disease progression, this may explain the negative outcome of the CANOPY-A study. Furthermore, in the CANOPY-1 trial, symptoms of lung cancer were reduced in the canakinumab arm. Should canakinumab mask the symptoms of lung cancer, it would be possible that, in CANTOS, canakinumab led to a delay in clinical presentation or potential identification of lung cancer as a cause of death.

Canakinumab was shown to be safe, and no unexpected safety signals were identified in CANOPY-A. IL-1β has been shown to be an important mediator of response to infections. Thus, IL-1β inhibition could have led to an increased risk of infections, which has been noted in the label of

inhibition could have led to an increased risk of infections, which has been noted in the label of canakinumab for autoinflammatory diseases.[14,26] Nevertheless, there were no significant differences in infection rates between the canakinumab and placebo treatment arms in CANOPY-A. On-treatment deaths due to infections and infestations were reported in <1% of patients across both arms.

In a retrospective study of 447,932 individuals in the UK Biobank, Swanton et al[10] showed that exposure to air pollution via particulate matter can drive the release of IL-1β, which can promote mutation-driven lung cancer development among never-smokers. The study also showed that anti–IL-1β therapy can prevent particulate matter–induced tumor formation in mouse models of lung cancer with *EGFR* and Kirsten rat sarcoma virus (*KRAS*) mutations. The authors further observed that *EGFR* and *KRAS* mutation drivers were, respectively, present in 15% and 53% of normal lung tissue from patients who had never smoked. As IL-1β may have a different role in the very early stage of the disease compared with after lung cancer is diagnosed, there may be potential for IL-1β inhibitors to prevent lung cancer development.

In conclusion, these results do not support the hypothesis that canakinumab would be beneficial as adjuvant treatment for patients with early-stage, resected NSCLC after cisplatin-based chemotherapy. However, the biomarker data from the CANOPY program are currently under analysis and may provide useful information on the most appropriate patient population for further investigation of IL-1β– targeted therapy. Finally, there may be a place for IL-1β inhibitor therapy in the prevention of lung cancer in high-risk individuals, which is being explored in the Can-Prevent-Lung (ClinicalTrials.gov identifier: NCT04789681) study.

Acknowledgment

Novartis employees are included as authors and were involved in the study design, data analysis and interpretation, writing of the report, and decision to submit the article for publication. The authors would like to thank the participating patients, their families, the study coinvestigators, and research contributors. David Demanse is thanked for his contribution. Medical writing assistance was provided by Alice Arcourt, PhD, of ArticulateScience Ltd.